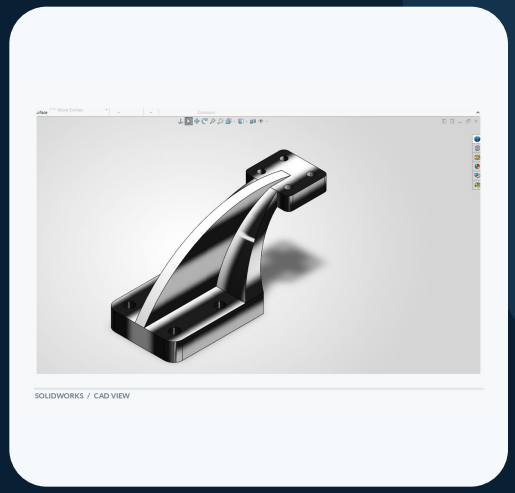
MECHANICAL DESIGN / SOLIDWORKS

Engineering portfolio.

Nine selected CAD modeling studies, documented with actual screenshots and PDF views.







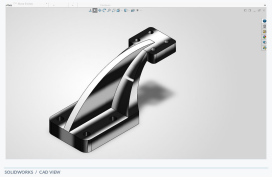
TH Deggendorf (THD)

Thesis application • CAD & mechanical design

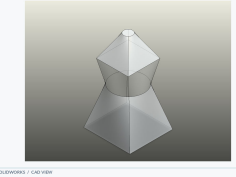
OVERVIEW

9 CAD studies

A quick look at the range of geometry, support components and housings.



Cantilever Bracket



Loft



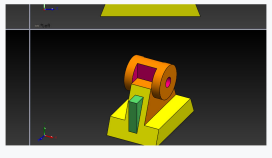
Multi-Port Valve



Pencil



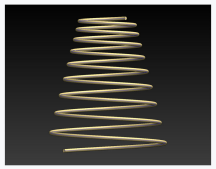
Pump Housing



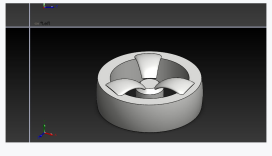
Shaft Support Bracket



Spoked Wheel Housing



Spring



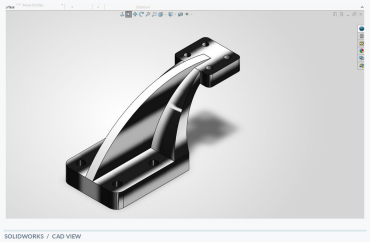
Wheel

SELECTED DESIGN STUDIES

Designing mechanical interfaces

A reinforcing bracket and a multi-port body with directional connections.

STRUCTURAL & SUPPORT COMPONENTS



Cantilever Bracket

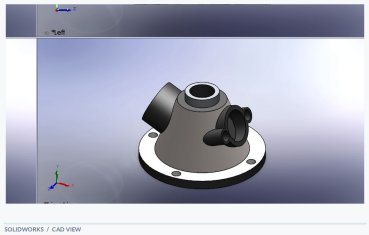
DESIGN IN FOCUS

A rib-reinforced cantilever bracket with a broad base and an elevated mounting interface.

VISIBLE FEATURES

Base plate with mounting openings • Elevated end mounting surface • Curved ribs connecting the two levels

FLUID & HOUSING CONCEPTS



Multi-Port Valve

DESIGN IN FOCUS

A multi-port valve body concept with a tapered main housing, inclined side connections and a mounting flange.

VISIBLE FEATURES

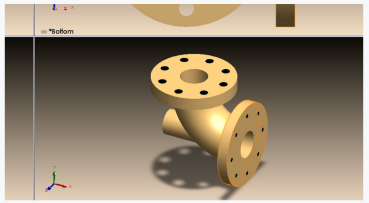
Tapered central body • Multiple angled cylindrical ports • Central opening and circular mounting flange

SELECTED DESIGN STUDIES

Geometry meets function

Flanged cylindrical interfaces and circular features with mounting lugs.

FLUID & HOUSING CONCEPTS



Pump Housing

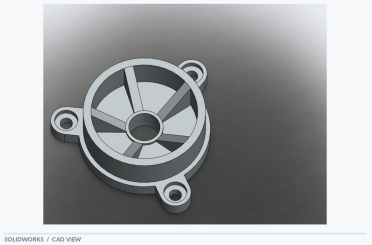
DESIGN IN FOCUS

A flanged pump-housing concept built around cylindrical passages and a side connection.

VISIBLE FEATURES

Cylindrical main housing • Circular flange and bolt-hole pattern • Perpendicular outlet or connection feature

ROTATIONAL & WHEEL COMPONENTS



Spoked Wheel Housing

DESIGN IN FOCUS

A spoked wheel-style housing with a ring, central hub and mounting lugs.

VISIBLE FEATURES

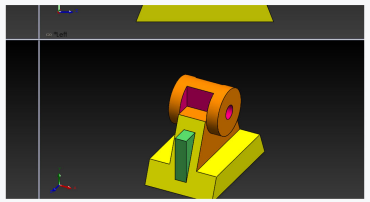
Radial spoke pattern and central opening • Circular outer ring • Three mounting lugs with through-holes

SELECTED DESIGN STUDIES

Supporting & shaping

A raised cylindrical support and helical geometry with changing diameter.

STRUCTURAL & SUPPORT COMPONENTS



Shaft Support Bracket

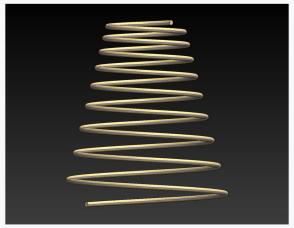
DESIGN IN FOCUS

A raised shaft-support bracket with a cylindrical upper feature and a wider base.

VISIBLE FEATURES

Raised cylindrical support feature • Upright central rib or web • Broad mounting base

ROTATIONAL & WHEEL COMPONENTS



Spring

DESIGN IN FOCUS

A tapered helical spring model with a gradually varying coil diameter.

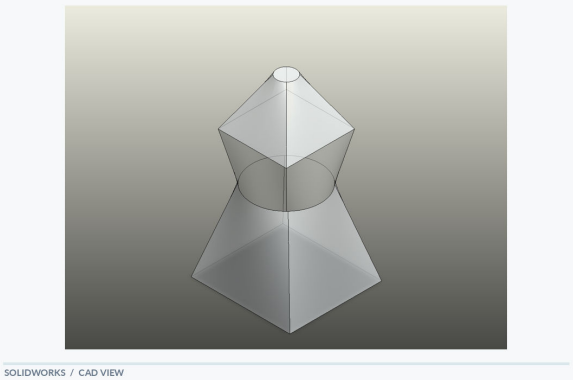
VISIBLE FEATURES

Multi-turn helical path • Larger lower and smaller upper coil diameters • Continuous circular wire profile

ADDITIONAL CAD STUDIES

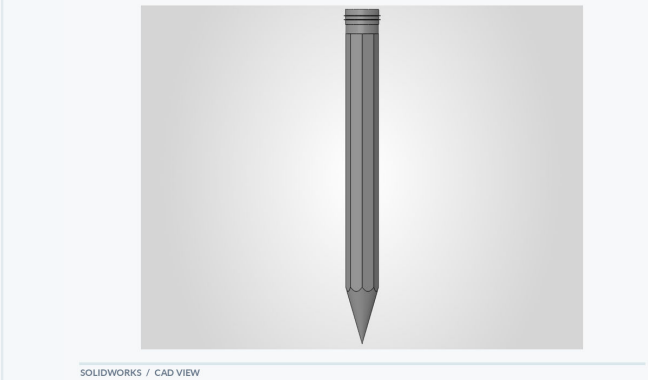
Breadth of modeling

Lofted geometry, an everyday object and a compact rotational component.



Loft

A symmetric loft study linking tapered polygonal sections through a cylindrical center.



Pencil

A slender pencil model combining a faceted main body with a tapered writing tip.



Wheel

A wheel-like rotational component with a circular rim and three internal radial elements.

FROM MODEL TO COMMUNICATION

What this portfolio shows

3D geometry	Part features, lofted forms, helical geometry and radial patterns.
Mechanical thinking	Mounting points, reinforcement, cylindrical interfaces and hubs.
Documentation	Consistent naming, original model evidence and clear design explanations.

View the full project archive:

https://github.com/PRASH5634/Solidworks_parts_drawings